Lesson 3.5 Guided Notes

Session 1 · Comparing Ratios with Common Terms

I can compare two ratios by scaling them until they share a common term. · 6.RP.A.3b

Word / Spanish	Meaning — complete the blank	Example
common multiple / múltiplo común	A number both terms can be _____ to.	2 and 3 both reach 6
least common multiple / mínimo común múltiplo	The _____ number that is a multiple of both numbers.	LCM of 2 and 3 is 6
stronger / más fuerte	More of the main ingredient for the _____ amount of water.	6 : 14 is stronger than 6 : 15

Key idea

To compare two ratios, scale them so one term is the _____ in both.

Then compare the _____ term. Less water for the same soap means a _____ mix.

Worked example

Mix A has 2 spoons of soap to 5 cups of water. Mix B has 3 spoons of soap to 7 cups of water. Which mix is stronger?

A common multiple of 2 and 3 is _____.

Mix A × 3 = 6 : _____. Mix B × 2 = 6 : _____.

Compare the water: 14 cups is less than 15 cups.

Answer: Mix _____ is stronger: less water for the same 6 spoons of soap.

You try

1. Tea A uses 2 bags for 9 cups of water. Tea B uses 3 bags for 12 cups. Scale both to 6 bags. Which tea is stronger?

_____

2. Scale each ratio so the first term is 20.
a. 4 : 7 = 20 : _____
b. 5 : 6 = 20 : _____

_____

Sum it up: To compare two ratios, I make ___ the same, and then I compare ___.

_____

Session 2 · Comparing Rates with Benchmarks

I can use a benchmark to compare two rates without dividing. · 6.RP.A.3b

Word / Spanish	Meaning — complete the blank	Example
benchmark / punto de referencia	A shared amount you scale _____ choices to.	12 lb for both stores
rate / tasa	Compares two amounts with _____ units.	$10 for 4 lb
unit rate / tasa unitaria	The amount for exactly _____.	$2.50 per lb

Key idea

Pick a benchmark that both amounts _____ into evenly, like 20 for 2 and 4.

Scale each rate to the benchmark, then _____ the results.

Worked example

Hose A fills 6 gallons in 2 minutes. Hose B fills 10 gallons in 4 minutes. Which hose fills faster?

Benchmark: _____ minutes for both hoses.

Hose A: 6 × 10 = _____ gallons. Hose B: 10 × 5 = _____ gallons.

Compare: 60 gallons > 50 gallons in the same 20 minutes.

Answer: Hose _____ fills faster.

You try

1. Store A: 3 lb for $8. Store B: 4 lb for $10. Use a 12-lb benchmark. Which is the better buy?

_____

2. Name a benchmark both amounts can reach.
a. 4 min and 6 min _____
b. 5 lb and 8 lb _____

_____

Sum it up: A benchmark helps me compare two rates because ___.

_____

Session 1 · Model, reason, and explain

Comparing Ratios with Common Terms · 6.RP.A.3b

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Mix A uses 2 cups of concentrate for 5 cups of water. Mix B uses 3 cups for 7 cups of water. Compare them using 6 cups of concentrate in each. Which tastes stronger if strength means more concentrate per cup of water?

Mix	Concentrate	Water
A	2	5
B	3	7

_____

_____

_____

Extend your reasoning

Could you compare using a common water amount instead? Show one valid comparison.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

Comparing Rates with Benchmarks · 6.RP.A.3b

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Hose A fills 8 gallons in 2 minutes; B fills 15 gallons in 5 minutes. Compare at 10 minutes and by unit rate. Explain why both methods agree.

Hose	Given rate	At 10 min
A	8 gal / 2 min	?
B	15 gal / 5 min	?

_____

_____

_____

Extend your reasoning

Give a different common-time benchmark and explain why it produces the same decision.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?